

Class 32: Orthogonal Sets

Definition: A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ in $\mathbf{R}^n$ is an orthogonal set if

Theorem: Any orthogonal set of nonzero vectors will be linearly independent.

Proof:

Any set of n orthogonal vectors in $\mathbf{R}^n$, will form an orthogonal basis for $\mathbf{R}^n$. The nice thing about an orthogonal basis is that it is very easy to find the coordinate vector relative to that basis.

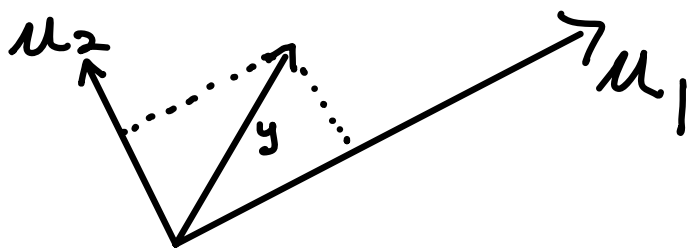
Theorem: If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthogonal basis for some subspace W in $\mathbf{R}^n$, then any vector $\mathbf{w}$ in W may be written as $c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k$ where each weight is computed by

Proof:

Example: Prove that the set $\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbf{R}^3$.

Write $\vec{w} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$

Let $\mathbf{u}_1$ and $\mathbf{u}_2$ be orthogonal vectors and let $\mathbf{y}$ be any vector in the span $\{\mathbf{u}_1, \mathbf{u}_2\}$



Problem #14 Let $\vec{y} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$. Write $\vec{y}$ as the sum of a vector in $\text{span}\{\vec{u}\}$ and a vector orthogonal to $\vec{u}$.

Big Question: Let W be a subspace of $\mathbf{R}^n$ and let $\mathbf{y}$ be any vector in $\mathbf{R}^n$. Which vector in W is closest to $\mathbf{y}$?

Strategy:

Step 1.

Step 3.

Two problems to explore:

1. How do you construct an orthogonal basis for W ?
2. How can you decompose $\mathbf{R}^n$ into W and $W^\perp$?